

**Group Design Project: Final Design Reports****Assigned: Monday, February 9, 2026****Due: Monday, March, 23 2026 via Canvas****(An unrecoverable penalty of 20% per calendar day may be assessed for late submissions)**

Your final assignment for your Group Design Project includes the following parts:

Part A) Design an ideal (*i.e.*, reversible) Rankine power cycle to produce the electrical power you specified for your Preliminary Design Reports using your conventional energy source (coal, natural gas, oil, or nuclear).

Part B) Correct any errors in your analysis for your Preliminary Design Reports and also explore strategies to improve/optimize the performance of the ideal Rankine power cycle from your preliminary design. In particular, to provide fair workloads for groups differing in number of team members:

- Groups of three: Explore one different strategies
- Groups of four: Explore two different strategies
- Groups of five: Explore three different strategies

Groups of three to five are required to include and submit work based on AspenPlus for at least one of their strategies. However, all groups are encouraged to use AspenPlus in implementing any strategies pursued.

Part C) Summarize the outcomes for your group project. Your summary should consist of:

- 1) A comparison, in the form of a table or tables, comparing among your designs—your preliminary design for an ideal cycle using your renewable/sustainable energy source (after you have made any corrections noted in feedback from your original submission), your design for an ideal cycle using your conventional energy source, your design for an air and liquid cooled process, and your set of improved/optimized designs for ideal cycles using your renewable/sustainable energy source and cooling method of choice—the following properties:

- Type & overall efficiency for cycle and mass flow rate(s) for working fluid.
- Type of energy source, required rate of heat transfer from the hot reservoir,  $\dot{Q}_H$ , and specifications for the supply of heat to the cycle (*i.e.*, mass rate at which fuel is burned, mass flow rate for geothermal or ocean water, mass flow rate for pressurized light water, or size of solar collector).
- Type of cooling, required heat transfer rate to the cooling water, & mass flow rate for cooling water

- 2) A brief statement (no more than 300 words) summarizing your general conclusions, what you have learned, and potential follow-ups for your project. *Note that concluding your current designs using your renewable/sustainable energy source are unfavorable is acceptable, justified, and proper if your results support this conclusion!*

Part D) Summarize in a brief statement (no more than 200 words) your *individual* specific contributions to the proposal (and its revision, if submitted), preliminary design report, and final design report. Also reflect what you *individually* have learned in this project and what you would consider doing further and/or differently.

## Specific Guidance for Individual Groups for Part A: Design of Cycles Based on a Conventional Energy Source

### Operating constraints

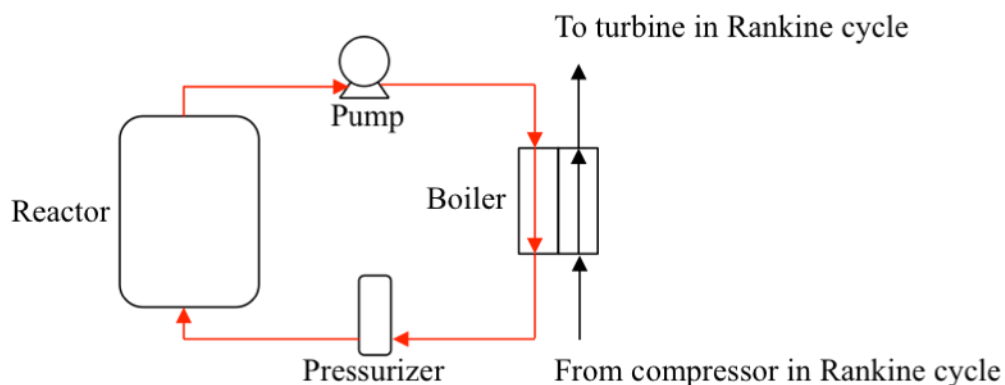
- Design an ideal Rankine/compression cooling cycle (*i.e.*, both turbine and compressor operate adiabatically & reversibly with isentropic efficiencies of 100%) using water as your working fluid to deliver the net power (in multiples of watts) specified for your project.
- The temperature of working fluid entering the condenser must be at least 3 °C higher than temperature of your entering cooling water.
- The difference in temperature between the cooling water exiting and entering the condenser must be no more than 2 °C.
- The pressure drop across the turbine must be at least 50 bar.

### Suggested operating conditions for your Rankine cycle using natural gas, coal, or oil as your conventional energy source

- If your group has not identified an estimate for the energy content of your conventional fuel, use  $\Delta\hat{h}_{comb} = 10.8 \text{ kWh/kg}$  for natural gas,  $\Delta\hat{h}_{comb} = 2.8 \text{ kWh/kg}$  for lignite coal, &  $\Delta\hat{h}_{comb} = 12.5 \text{ kWh/kg}$  for oil (from [http://www.mpoweruk.com/energy\\_resources.htm](http://www.mpoweruk.com/energy_resources.htm)).

### Use of a pressurized light water nuclear reactor to heat boiler and suggested operating conditions

- The simplest working model for a nuclear power plant is based on a pressurized light water nuclear reactor with a pair of coupled flow loops. The primary loop (depicted below) consists of the reactor and a pump and couples to a secondary loop (which forms the Rankine cycle) through a counter-current heat exchanger functioning as the boiler on the Rankine secondary loop. In a pressurized water reactor the fluid in the primary loop is water at high pressures (*e.g.*, 152 bar), depicted as red streams below; water as the working fluid in the secondary loop formed by the Rankine cycle is depicted as black streams.



- As an approximation we will ignore the contributions of the pressurizer and pump to the energy balance for the primary loop and assume the outlet from the reactor and inlet to the boiler in the primary loop are at  $T_{reactor\ outlet} = 324\text{ °C}$  and the outlet from the boiler in the primary loop and inlet to the reactor are at  $T_{reactor\ inlet} = 288\text{ °C}$ .

- Apply the relationship  $\dot{Q}_H = -\dot{Q}_{\text{boiler, primary}} = \dot{m}_{\text{primary}} (\hat{h}_{\text{reactor outlet}} - \hat{h}_{\text{reactor inlet}})$ , where  $\dot{m}_{\text{primary}}$  is the mass flow rate in the primary loop and  $\hat{h}_{\text{reactor outlet}}$  &  $\hat{h}_{\text{Reactor inlet}}$  the specific enthalpies of the reactor outlet and inlet, respectively.
- Note that Wolfram Alpha provides a platform for estimating the specific enthalpies of subcooled liquids at pressures above what Koretsky provides.

#### Questions to be addressed for Part A

For your Rankine/Cooling cycle using your conventional energy source:

- 1) Complete the design table, identifying the properties of each of the four states in the cycle. Explain your rationale for how you specified or identified the temperature and pressure for each state.
- 2) Determine for an ideal cycle (*i.e.*, assuming isentropic efficiencies are 100% for both the turbine & compressor):
  - a) The mass flow rate for the working fluid,  $\dot{m}$  (in kg/s)
  - b) The required heat transfer rate from the hot reservoir,  $\dot{Q}_H$  (in same units as power), and specifications for the supply of heat to the cycle:
    - Natural gas, oil, & coal: the mass rate (in kg/s) at which your fuel is burned. A good practice also is to verify that the adiabatic flame temperature for your fuel is higher than the temperature of the working fluid exiting the boiler.
    - Nuclear: the mass flow rate (in kg/s) of pressurized light water in the primary loop.
  - c) The required heat transfer rate to the cold reservoir,  $\dot{Q}_C$  (in same units as power), and the required mass flow rate (in kg/s) of cooling water. Make sure to state explicitly the inlet temperature and temperature rise for your cooling water, considering that heat transfer must be from the working fluid to the cooling water.
  - d) The efficiency for the cycle,  $\eta$ .

#### Specific Guidance for Part B: Improvement/Optimization of Rankine/Cooling Cycles Based on a Renewable/ Sustainable Energy Source

Your primary goal for this part is to identify how you can increase the efficiency of the ideal (reversible) cycle for your preliminary design. Secondary goals include decreasing the magnitudes of the rates for heat transfer from your hot reservoir and to the cooling water, such that lower mass rates of consumption of your energy source (or smaller collector size for a solar thermal process) and lower mass flow rates for cooling water can be used, and increasing the quality of the outlet from the turbine (to prolong the working life of the turbine by reducing the amount of liquid passing through it). In general, you can improve the performance of a power cycle by increasing the average temperature during heat addition to the feed to the boiler—termed the feedwater—and decreasing the average temperature during heat rejection in the condenser. Note that you should focus solely on ideal cycles in these designs.

The following types of general strategies can be considered:

- Option 1) Vary the operating conditions for your cycle. In particular, improved cycle performance—in terms of increased overall efficiency and smaller magnitudes for rates of heat transfer—typically results from increasing the pressure and

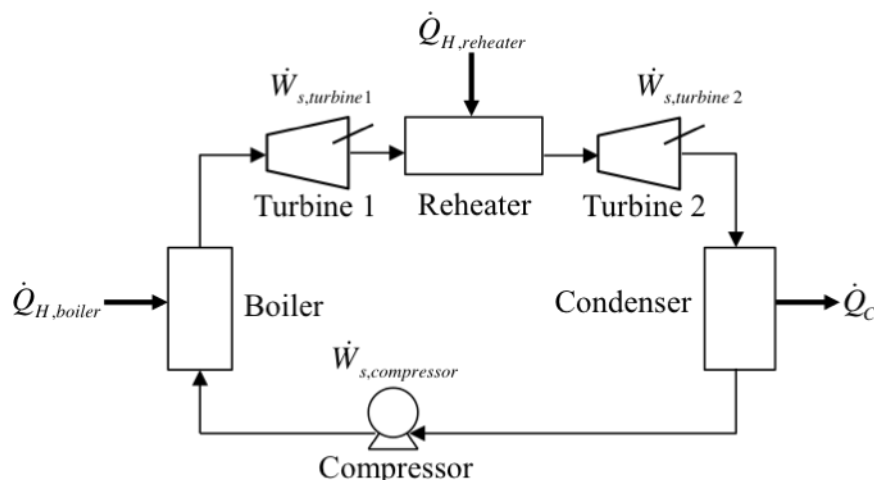
temperature drops across the turbine. In fact, current designs for Rankine cycles for power plants operated with fossil or nuclear fuels commonly are based on a supercritical Rankine cycle in which State 1 is a supercritical fluid. Alternatively, a Brayton cycle is similar to a Rankine cycle except each state in the cycle is a superheated vapor.

Option 2) Change the working fluid. This strategy is particularly useful for cycles with working fluids other than water.

Option 3) Change the configuration for your cycle by introducing a reheater and a second turbine between the turbine and condenser, as depicted below. Heat at a rate

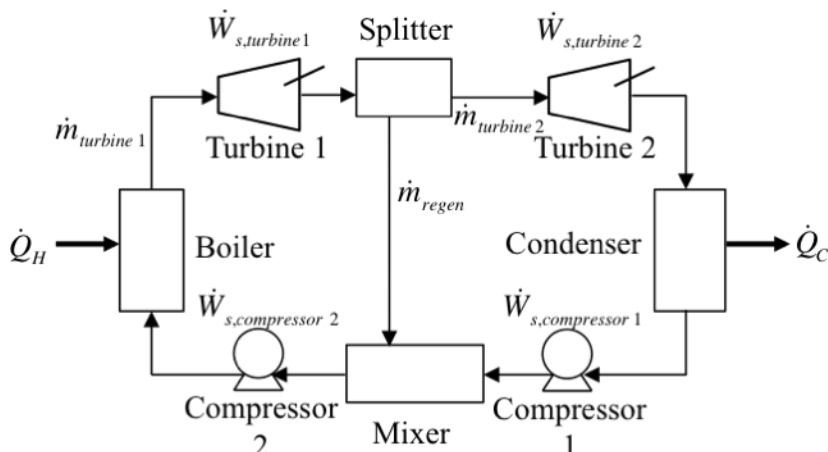
$\dot{Q}_{H, reheater}$  is transferred to the working fluid in the reheater, such that

$\dot{Q}_H = \dot{Q}_{H, boiler} + \dot{Q}_{H, reheater}$  &  $\dot{W}_{s, net} = \dot{W}_{s, turbine1} + \dot{W}_{s, turbine2} + \dot{W}_{s, compressor}$ . Preferences for the outlet for the turbines are as discussed previously. For this modified configuration label the states for the working fluid sequentially starting with State 1 for the inlet to the first turbine and modify your design table to include information on the inlet and outlet for the reheater.



Option 4) Change the configuration for your cycle by increasing the temperature for heat addition using a regenerative Rankine cycle with an open feedwater heater, configured as depicted on the top of the next page. In such a cycle superheated vapor is pulled from the power-producing side of the cycle to pre-heat the feedwater fed to the boiler. In the variation proposed here, two turbines, two compressors, a splitter, and a mixer are used:

- A high-energy superheated vapor is expanded in the first turbine to a lower-energy superheated vapor;
- The splitter separates the product of this turbine into two streams, one fed to a second turbine and a second fed to a regenerator termed an open feedwater heater;
- The second turbine further expands its feed to the condenser pressure;
- The first compressor produces a subcooled liquid at the feedwater pressure by increasing the pressure of the saturated liquid produced by the condenser;



- The mixer acting as a regenerator combines the feed from the splitter with the product of the first compressor to produce a saturated liquid at the feedwater pressure;
- The second compressor increases the pressure of its saturated-liquid feed to produce a subcooled liquid fed to the boiler.

This design requires specifying the split after the first turbine, the outlet pressure and temperature from the first turbine, and the feedwater pressure, with

$$\dot{W}_{s,net} = \dot{W}_{s,turbine1} + \dot{W}_{s,turbine2} + \dot{W}_{s,compressor1} + \dot{W}_{s,compressor2} \text{ \& } \dot{m}_{turbine1} = \dot{m}_{turbine2} + \dot{m}_{regen}.$$

For this modified configuration label the states for the working fluid sequentially starting with State 1 for the inlet to the first turbine and modify your design table to include information on the inlet and outlets for the splitter, the second turbine, the mixer, and the second compressor. Further information on regenerative Rankine cycles is available at [https://www.ecourses.ou.edu/cgi-bin/ebook.cgi?doc=&topic=th&chap\\_sec=10.3&page=theory](https://www.ecourses.ou.edu/cgi-bin/ebook.cgi?doc=&topic=th&chap_sec=10.3&page=theory).

Combinations of these general strategies (*e.g.*, changing both the working fluid and operating conditions) can be particularly useful to increase performance! Make sure to explain your rationale for how your identified key temperature and pressures.

Consider that once you have a flowsheet for a reversible Rankine cycle configured using AspenPlus, it is straightforward and probably less time-consuming to manipulate operating conditions and change the working fluid using Aspen than doing so manually using tabulated data. Also, modifying both the configuration and operation of the cycle, such including a second turbine and reheater (specify the outlet pressure and either temperature or vapor fraction for the reheater) or introducing regeneration (specify the flow rate to each turbine, the outlet conditions from the first turbine, and the outlet pressure for the mixer), may be done relatively easily compared to performing the necessary calculations by hand.

Questions to be addressed for Part B

For each strategy in your improved/optimized design of your ideal cycle from your preliminary design:

- 1) Complete the design table for an ideal cycle, identifying the properties of each of the states/streams in the cycle.
- 2) Determine:
  - a) The mass flow rate for the stream exiting the boiler,  $\dot{m}$  (in kg/s); for regenerative cycles also specify the mass flow rates (in kg/s) for the streams exiting the splitter.
  - b) The required heat transfer rate from the hot reservoir,  $\dot{Q}_H$  (in same units as power), and specifications for the supply of heat to the cycle:
    - Biomass & oil: the mass rate (in kg/s) at which your fuel is burned. A good practice also is to verify that the adiabatic flame temperature for your fuel is higher than the temperature of the working fluid exiting the boiler.
    - Geothermal: the mass flow rate (in kg/s) required from your geothermal source.
    - Ocean thermal: the mass flow rate (in kg/s) at which warm water is supplied to boiler.
    - Solar thermal: the size (in m<sup>2</sup> and football fields) for the solar collector.
  - c) The required heat transfer rate to the cold reservoir,  $\dot{Q}_C$  (in same units as power), and the required mass flow rate (in kg/s) of cooling water. State explicitly the inlet temperature and temperature rise for your cooling water.
  - d) The efficiency for the cycle,  $\eta$ .

**Guidelines for Submission**Submission for Parts A, B & C

- 1) One member should submit for the group a single PDF with their combined work for Parts A, B (including preliminary design, if revised), and C.
- 2) Include a cover page with your project name, the names of each team member alphabetized based on last name, the course name, and date of submission.
- 3) Include sufficient details for the reader to easily verify your outcomes.
- 4) Scanned handwritten work with legible script is permissible.
- 5) There is no page limit. Use 1" margins on all sides, and place a page number at the bottom of each page after the cover page.

Submission for Part D

Each group member should submit *separately* a PDF with his/her personal self-reflection.

**Rubric for Assessment**

Please review the rubric with detailed breakdown of criteria for assessment for Parts A-D below before making your submissions to confirm key elements sought. Revised proposals will be assessed with the criteria stated in the assignment for proposals.

**Part A—Design for Ideal Cycle for Conventional Source**

Completed design table for each state for:

| $P$ | $T$ | $\hat{h}$ or $h$ | $\hat{s}$ or $s$ | $x_2$ (or $f_{v,2}$ ) |
|-----|-----|------------------|------------------|-----------------------|
|-----|-----|------------------|------------------|-----------------------|

Turbine is isentropic

Compressor is isentropic

Identified inlet temperature of cooling water

Satisfied constraints on State 2

Satisfied constraints on State 1

Determined correctly mass flow rate for working fluid

Determined correctly  $\dot{Q}_H$  for cycle

Determined correctly  $\dot{Q}_C$  for cycle

Determined correctly specifications for energy source

Determined correctly mass flow rate for cooling water

Determined correctly overall efficiency for cycle

**Part B—Design for Improved Cycle for Renewable Source (for each strategy)**

Indicated explicitly proposed strategy for improvement compared to preliminary design

Completed design table for each state for:

| $P$ | $T$ | $\hat{h}$ or $h$ | $\hat{s}$ or $s$ | $x_2$ (or $f_{v,2}$ ) |
|-----|-----|------------------|------------------|-----------------------|
|-----|-----|------------------|------------------|-----------------------|

Turbine is isentropic

Compressor is isentropic

Satisfied constraints on State 2

Satisfied constraints on State 1

Determined correctly mass flow rate(s) for working fluid

Determined correctly  $\dot{Q}_H$  for cycle

Determined correctly  $\dot{Q}_C$  for cycle

Determined correctly specifications for energy source

Determined correctly mass flow rate for cooling water

Determined correctly overall efficiency for cycle

Part C—Project Summary

- 1) Summary table(s)  
Summary table(s) identified for each design:
  - Type of energy source
  - Type of cycle
  - Cycle efficiency
  - Mass flow rate(s) for working fluid
  - Specifications for supply of heat to cycle
  - $\dot{Q}_H$                        $\dot{Q}_C$
  - Mass flow rate for cooling waterSummary table(s) organized
- 2) General conclusions
  - Arrived at a set of conclusions
  - Conclusions justified by results
  - Conclusions reflect understanding of thermodynamics
  - Conclusions communicated effectively & clearly
- 3) Learning/future directions/changes
  - Identified at least three items learned
  - Learning reflect understanding of thermodynamics
  - Identified at least two ideas for further/different consideration
  - Learning/future directions/changes communicated effectively & clearly

Part D—Self-Reflection

- Attempted self-reflection
- Clearly identified his/her contributions
- Clearly identified what learned individually
- Clearly identified what would do further/differently
- Communicated effectively & clearly
- Shared group work effectively

Assessment: Scores of Meets Expectations will be earned with no more than 2 moderate deficiencies, with all work being easy to follow/understand. Scores of Excellent can be achieved with no more than two minor deficiencies and a polished quality of work.

A minor deficiency for example might be a math/calculator error or misreading a number from a table. A moderate deficiency for example might be a value that is missing or more significant math errors involving mishandling of units, etc.

A missing or otherwise incomplete self-assessment will result in a loss of one point on the final report grade.